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Education

Sahyadri College of Engineering and Management, Mangaluru
B.E. in Computer Science (Artificial Intelligence & Machine Learning)

Expected 2027
CGPA: 9.46

Research and Innovation

Multimodal Feature Level Fusion for Arecanut Quality Assessment Submitted to Elsevier *Array* Journal / SSRN Preprint [SSRN]

Developed a multimodal RGB-X-ray deep learning pipeline for arecanut quality classification, achieving 99.80% accuracy using CNN-based architectures with PyTorch and TensorFlow/Keras. Evaluated using cross-validation and standard classification metrics.

Empowering Wildlife Conservation Through Real-Time Anti-Poaching System
DISCOVER 2025 [DOI]

Published, IEEE

Co-authored a real-time IoT and AI based anti-poaching system using YOLOv8n for intrusion detection in wildlife reserves, with SMS-based alerting and explainable AI for forest official use.

Projects

RxSignal - Drug Interaction Risk Intelligence Platform XGBoost, PyTorch, FastAPI, React [GitHub]

- Built an end-to-end drug safety platform ingesting 1.06B FDA adverse event records (FAERS 2025) with a chunked DuckDB pipeline, computing PRR/ROR signals across 43.8M drug-event triples.
- Trained XGBoost (90.6% accuracy, 99.35% recall) and a GraphSAGE GNN on a 13,160-node drug graph; served predictions via FastAPI and React in ~1ms using pre-computed embeddings.

AESCULAPIUS - Self-Healing Multi-Agent RAG Platform LangGraph, Gemini/Ollama [GitHub]

- Built a self-healing multi-agent RAG platform that retrieves, reranks, verifies, and autonomously improves responses using an 8-agent LangGraph workflow.
- Integrated Pinecone, BAAI embeddings, cross-encoder reranking, hallucination detection, confidence scoring, and FastAPI with a React-based interface.

EEG-Based Early Detection of Neurodegenerative Disorders Python, TensorFlow, React, FastAPI [GitHub]

- Built a CNN-BiLSTM and GCN deep learning pipeline on 19-channel EEG signals to classify Alzheimer's, FTD, and Cognitively Normal with 96.74% accuracy.
- Developed a FastAPI backend for EEG preprocessing with a React frontend for brain connectivity visualization.

Skills

Programming Languages: Python, SQL, Java, C/C++, HTML/CSS, JavaScript

AI/ML & Development: Machine Learning, Deep Learning, Feature Engineering, Model Training & Evaluation, Predictive Modeling, Data Visualization, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, React, Node.js, Express.js, MERN Stack, MongoDB, LangChain, LangGraph

Tools & Platforms: Git, GitHub, Jupyter, Docker, Vercel, MongoDB Atlas, Postman, n8n, Power BI, Figma, Miro, Canva

Soft Skills: Problem-Solving, Strategic Thinking, Collaboration, Leadership, Mentoring

Achievements

RIDA – IEEE YESIST12 Special Track Finalist, [April 2026]

Presented a clinical decision support tool using a biomedical knowledge graph and GraphSAGE GNN to return ranked rare-disease diagnoses, aimed at reducing diagnostic delay for under-resourced GPs.

StreamSentinel – IEEE YESIST12 WePOWER Track Finalist, [April 2026]

Presented a federated-learning IoT platform for real-time water quality anomaly detection in rural communities, held at Manipal Institute of Technology, Bengaluru.

VEDHA – VVCE EdTech Hackathon Finalist, Nov 2024

Built Edu-Ease, an assistive learning app for special-needs children with AI-based assessments, text-to-speech and speech-to-text, and adaptive content; selected as finalist among national entries.

Chair, IEEE Student Branch & Computational Intelligence Society 2025 – Present

Coordinate technical events and student participation in competitions.

LeetCode 300+ Problems Solved [Link]

Solved 300+ DSA problems covering arrays, trees, graphs, and linked lists.

Certifications

Machine Learning with TensorFlow Infosys Springboard [Certificate]

Master Computer Vision™ OpenCV3 in Python and Machine Learning Infosys Springboard [Certificate]

Python for Data Science Infosys Springboard [Certificate]